

Haolin Chen

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SUMMARY

Research Scientist specializing in **Large Language Model (LLM) training, reinforcement learning, and agentic AI systems**. Experience in leading **end-to-end LLM research**, including pre-training, diffusion-based modeling, and reasoning. Expert in PyTorch, transformers, TRL, veRL, TPU/GPU training, and large-scale data synthesis.

WORK EXPERIENCE

Head of Research

actAVA AI

Pleasanton, CA

March 2026 – Present

Senior Applied Scientist

Salesforce AI Research

Palo Alto, CA

August 2025 – February 2026

Applied Scientist

Salesforce AI Research

Palo Alto, CA

January 2024 – July 2025

- Lead LLM reasoning research. Developed **LaTRO**, a reinforcement learning-based algorithm to train LLMs to reason during post-training. The algorithm shows at most **~20% performance gain** on downstream math tasks.
- Lead diffusion language model research, in which LLMs are trained using discrete diffusion. Develop **CoDA-1.7B**, a light-weight, fast diffusion language model for coding that has **performance matching 7B models with 4x efficiency**.
- Lead RL data synthesis research, developed **Webscale-RL**, an open-source data synthesis pipeline that creates high quality RL datasets **at the scale of pre-training data**. Open-sourced a **1 million** dataset for RL.
- Developed the **xLAM** series, a foundational LLM family (**#1 on BFCL leaderboard**) for function calling and agentic environments; showcased demo for travel planning using xLAM at ICLR 2025.
- Prototyping LLM agents for: hierarchical planning, sales pitching, and customer service. Developed **multi-turn agent evaluation** systems with **APIGen-MT**, a multi-turn, agentic data synthesis framework.

Data Scientist

Outreach

Seattle, WA

August 2022 – September 2023

- Built PySpark-based predictive tree models for opportunity win rates, improving F1 by **10%** through feature optimization.
- Integrated GPT-3.5 with RAG to summarize deal histories, improving sales insights and automation.

Data Scientist Intern

Outreach

Seattle, WA

July 2021 – Dec 2021

- Developed a transformer-based ML system for market research and organization clustering. Outperform the existing rule-based baseline by **7x** in precision, and **3x** in F1 score.

EDUCATION

University of California, Davis

Ph.D. in Applied Mathematics. Advisor: Luis Rademacher.

Davis, CA

Sep. 2017 – July 2022

Nankai University

Bachelor of Science in Physics and Bachelor of Science in Mathematics.

Tianjin, China

Sep. 2012 – Jun. 2017

Full list:  Google Scholar]

Artificial Intelligence

- **Cura 1T: Specialized Model for Agentic Healthcare.** arXiv preprint arXiv:2607.15314.
[Chen, H.](#), Qi, L., Brown, S., Metelski, D., Xia, T., Lee, J., Wang, Q., Riley, K., Wang, F., & Yao, W.
 **Paper**
- **χ -Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?** arXiv preprint arXiv:2605.16679.
[Chen, H.](#), Metelski, D., Qi, L., Xia, T., Lee, J., Brown, S., Riley, K., Wang, F., et al.
 **Paper**
- **CoDA: Coding LM via Diffusion Adaptation.** arXiv preprint arXiv:2510.03270.
[Chen, H.](#), Wang, S., Qin, C., Pang, B., Liu, Z., Qiu, J., Zhang, J., Zhou, Y., Chen, Z., Xu, R., Heinecke, S., Savarese, S., Xiong, C., Wang, H., Yao, W.
 **GitHub:** 58 ★
- **Webscale-RL: Automated Data Pipeline for Scaling RL Data to Pretraining Levels.** ICLR 2026.
Cen, Z., [Chen, H.](#), Wang, S., Liu, Z., Liu, Z., Zhao, D., ... & Yao, W.
 **GitHub:** 72 ★
- **Language models are hidden reasoners: Unlocking latent reasoning capabilities via self-rewarding.** arXiv:2411.04282
[Chen, H.](#), Feng, Y., Liu, Z., Yao, W., Prabhakar, A., Heinecke, S., ... & Wang, H.
 **GitHub:** 123 ★
- **xLAM: A Family of Large Action Models to Empower AI Agent Systems.** arXiv preprint arXiv:2409.03215.
Zhang, J., Lan, T., Zhu, M., Liu, Z., Hoang, T., Kokane, S., Yao, W., Tan, J., Prabhakar, A., [Chen, H.](#), Liu, Z., ... & Xiong, C.
 **Paper** |  **GitHub:** 601 ★
- **Apigen-mt: Agentic pipeline for multi-turn data generation via simulated agent-human interplay.** NeurIPS 2025 Datasets & Benchmarks Track, to appear
Prabhakar, A., Liu, Z., Zhu, M., Zhang, J., Awalganekar, T., Wang, S., Liu, Z., [Chen, H.](#) ... & Xiong, C. (2025).

Patents

- **Systems and Methods for Generative Language Model Reasoning Process Optimization.** U.S. Patent Application Publication No. US 2026/0093997 A1; pending U.S. patent application, published April 2, 2026.
[Chen, H.](#), Feng, Y., Prabhakar, A., Liu, Z., Yao, W., Ho, R., Mui, L., Savarese, S., Wang, H., & Xiong, C.
 **Patent Application**

Machine Learning Theory

- **Overcomplete order-3 tensor decomposition, blind deconvolution and Gaussian mixture models,** SIAM Journal on Mathematics of Data Science 2022 4:1, 336-361.
[Haolin Chen](#), Luis Rademacher.

Machine Learning in Healthcare

- **The Research Landscape of Multiple Endocrine Neoplasia Type 1 (2000–2021): A Bibliometric Analysis.** Frontiers in Medicine, 1007.
Feng, C., [Chen, H.](#), Huang, L., Feng, Y., & Chang, S. (2022).
- **Trends in worldwide research in inflammatory bowel disease over the period 2012–2021: A bibliometric study.** Frontiers in Medicine, 1389.
Li, K., Feng, C., [Chen, H.](#), Feng, Y., & Li, J. (2022).
- **A Bibliometric Analysis of the Landscape of Parathyroid Carcinoma Research Based on the PubMed (2000–2021).** Frontiers in Oncology, 12, 298.
Feng, C., Tian, C., Huang, L., [Chen, H.](#), Feng, Y., & Chang, S. (2022).
- **Exploring the research landscape of the past, present, and future of thyroid nodules.** Frontiers in Medicine, 9.
Chen, P., Feng, C., Huang, L., [Chen, H.](#), Feng, Y., & Chang, S. (2022).

OTHER PROFESSIONAL EXPERIENCE

Reviewer

- NeurIPS 2025
- NeurIPS 2026
- Association for Computational Linguistics
- Mathematical Reviews/MathSciNet
- Journal of Computer Science and Technology